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Results of Brocard-Ramanujan problem on diophantine equation $n! + 1 = m^{2*}$

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Abstract

The Brocard-Ramanujan problem pertaining to the diophantine equation $n! + 1 = m^2$, a famously unsolved problem, deals with finding the integer solutions to the equation. Nobody has discovered any new solution of the problem beyond $n = 4,\ 5$ and 7 although many of us have tried it. Bruce Berndt and William Galway [10] had not found any new solution in 2000 by extensive computer search for a solution with n up to 10^9. The purpose of this study is to show that the solutions should satisfy some necessary and/or sufficient conditions. If $\sqrt{n!} = k+\epsilon,\ n > 1,\ 0 < \epsilon < 1$; then it has solution if and only if $n! = k(k+2)$ and $\epsilon,\ k$ are strictly monotonic increasing. It has only finitely many solutions which is not based on any conjecture or previous research on the Brocard-Ramanujan problem. For the new solution of Brocard- Ramanujan problem ($n \geq 10^5$), the value of ϵ should be more than $0.999\cdots905915$ (digit 0 is coming after 228287 numbers of 9 digit, which takes more than 66 pages in (LibreOffice Writer) indicating almost impossibility of new solution. If we consider $n \geq 10^9$, I am unable to calculate the said numbers of 9 digit in the value of ϵ in my personal laptop (with 8GB Ram) using MATHEMATICA 8. Finally, it has been claimed to discover that the problem has no further solution.

Keywords: Brocard-Ramanujan problem; Brocard' problem; $n! + 1 = m^2$*; Diophantine equation.*

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1 Introduction

The Brocard-Ramanujan problem was reported by Henri Brocard [1, 2] in the two written articles in 1876 and 1885 asking to search positive integers n and m satisfying the equation $n!+1 = m^2$, where $n!$ is the factorial. Later in 1913, unaware of Brocard's query, S. Ramanujan [3] reported the same problem in the following composition: "The number $n! + 1$ is a perfect square for the values 4, 5, 7 of n. Find other values."

It is believed that the only solutions of the problem are $(m, n) = (5, 4), (11, 5), (71, 7)$. The solutions of the problem in the pair form (m, n) is known as Brown's numbers. As reported by A. Gérardin [4] with arguments in 1906, the problem has no solutions for $7 < n < 25$ and m must have at least 20 digits if $m > 71$ for the solution. The computations of $n!$ up to $n = 63$ carried out by Hansraj Gupta [5] in 1935 provided no further solutions of the problem. Considering more general diophantine equations $x^k \pm y^k = n!$ and $m! \pm n! = x^k$, P. Erdős and R. Obláth [6] reported that most of these have only finitely many solutions. For example, the equation $x^k - 1 = n!$ has no solutions if $k > 1$ and possible exception of $k = 2$ and $k = 4$. They proved that the number of solutions is finite for the case of $k = 4$ i.e. for the equation $x^4 - 1 = n!$. Pollack and Shapiro [7] finally showed that the equation $x^4 - 1 = n!$ has actually no solutions.

In 1993, Marius Overholt [8] proved that "a weak form of the unproved diophantine inequality known as Szpiro's conjecture would imply that there are only finitely many solutions" of the problem. The weak form of Szpiro's conjecture, a special case of the ABC conjecture, can be defined as follows: Let $N_0(n) = \prod_{p|n} p$, in which p indicates a prime; a, b, and c denote positive integers together with relatively prime in pairs and follow the equation $a+b = c$. Then the weak form of Szpiro's conjecture implies that there exists a constant k such that $|abc| \leq [N_0(abc)]^k$.

In 1996, A. Dabrowski [9] carried out the extension of the result, under the identical condition, to all equations of the type $n! + A = m^2$ in terms of an arbitrary natural number A. Under the identical condition or even a harder constitution, the result of Overholt was further carried forward by several researchers [11, 13, 14] to equations of the form $n! = P(m)$ or even $n(n-1)\cdots(n-k+1) = P(m)$, in which P is a polynomial of degree ≥ 2 with integer coefficients, and $k,\ m,\ n$ are integers with $1 \leq k \leq n$.

In 2000, Bruce Berndt and William Galway [10] found no solution of the problem based on computations up to $n = 10^9$, reported a relation between Legendre's symbol $(\frac{a}{p})$ and the solutions of $n! + 1 = m^2$ indicating $(\frac{n!+1}{p}) = 1$ or 0.

On the basis of consideration of Szpiro's conjecture, Dufour and Kihel [12] in 2004 proved

that Hall's conjecture weak form implies that $n! + A = m^2$ (if the integer A is not a q-th power of an integer) can only have finitely many solutions, where Hall's conjecture defines a special form of the abc-conjecture.

In 2008, Kihel and Luca [15] presented some variants of the Brocard-Ramanujan problem considering the form $x^p \pm y^p = \prod_{k=1\ (k\nmid n)}^{n} k$ and showed that there are only finitely many integer solutions $(x,\ y,\ p,\ n)$ if $p\ (\geq 3)$ a prime number and $\gcd(x, y) = 1$.

In 2013, Liu [16] wrote a Master of Science thesis on Brocard's problem and its variations in Waikato University on the basis of reviews for the old methods in order to solve various forms of the Brocard's problem and presented some proofs in his own way. Concerning variations of Brocard's problem, Maciej Ulas [18] considered the equation $n! + A = m^2$ and proved that there were infinitely many A's such that the equation $n! + A = m^2$ has at least three solutions in natural number. Dabrowski and Ulas [19] investigated Diophantine equations of the form $m^2 = px_n + q$, in which p, q are fixed integers, $x_n = g(1)g(2)\cdots g(n)$ and $g : \mathbb{N}_+ \to \mathbb{N}_+$ is an increasing function. They proved that the equations had at least four solutions in positive integers m and n.

Maximum number of the results of the Brocard-Ramanujan problem and its variations has been derived by using the ABC conjecture or its variants [19]. A question is arising naturally that how the results can be proved without the help of this assumption [19]? In view of the above, one of the purpose of this study is to show the finiteness of the number of solutions of $n! + 1 = m^2$, which is not based on any conjecture or previous study and the solutions should satisfy some necessary and/or sufficient conditions. The proof of the existence of new solution of equation (1) seems to be as distant as the establishment of the nonexistence of odd perfect numbers and the two problems although not equivalent are not dissimilar.

We are tempted to believe the stronger statement that Brocard–Ramanujan problem has no other solutions except $(m, n) = (5, 4),\ (11, 5),\ (71, 7)$; however, it can be claimed that this study have been able to establish this stronger statement. Moreover, results on Brocard–Ramanujan problem show that if $\sqrt{n!} = k + \epsilon,\ n > 1,\ 0 < \epsilon < 1$; then it has solution if and only if $n! = k(k + 2)$ and $\epsilon,\ k$ are strictly monotonic increasing. For new solution of Brocard-Ramanujan problem considering $n \geq 10^5$, the value of ϵ should be more than $0.999\cdots905915$ (digit 0 is coming after 228287 numbers of 9 digit, which takes more than 66 pages in LibreOffice Writer) concluding the impossibility of new solution. I am unable to calculate the numbers of said 9 digit in the value of ϵ in my personal laptop (with 8GB Ram) using MATHEMATICA 8

when $n \geq 10^9$.

2 Results and Discussion

2.1 Theorem

The natural number m will be a solution of the diophantine equation

$$n! + 1 = m^2;\ n, m \in \mathbb{N} \tag{1}$$

if and only if $m = [\sqrt{n!}] + 1$, where for $x \in \mathbb{R}$, $[x]$ indicates greatest integer function. Moreover for the solution of equation (1), we can express $n!$ as $n! = 2^{\beta_1} \times$ odd number $= (2a) \times (2^{\beta_1 - 1}b)$ with a and b relatively odd natural numbers satisfying $\gcd(a, 2^{\beta_1 - 1}b) = 1$ and the difference of $2a$ and $2^{\beta_1 - 1}b$ will be 2 with $\beta_1 = [\frac{n}{2}] + [\frac{n}{4}] + [\frac{n}{8}] + \cdots$.

Proof: The equation (1) can be written as $n! = m^2 - 1 = (m-1)(m+1)$.

Then we can express it in the form:

$$n! = [\sqrt{n!}]([\sqrt{n!}] + 2) \tag{2}$$

Or, $[\sqrt{n!}]^2 + 2[\sqrt{n!}] - n! = 0$.
Or, $[\sqrt{n!}] = \frac{-2 \pm \sqrt{4+4n!}}{2} = -1 \pm \sqrt{1+n!}$.
So, $[\sqrt{n!}] = \sqrt{1+n!} - 1$.

$$\text{Or, } [\sqrt{n!}] + 1 = \sqrt{1+n!} = \sqrt{m^2} = m \tag{3}$$

Thus if m is a solution of equation (1), then it will be in the form $m = [\sqrt{n!}] + 1$. Hence the condition is necessary.

Conversely, let $m = [\sqrt{n!}] + 1$. Then $(m-1)(m+1) = [\sqrt{n!}]([\sqrt{n!}] + 2)$.
Or, $n! = m^2 - 1 = [\sqrt{n!}]([\sqrt{n!}] + 2)$ i.e. $n! + 1 = ([\sqrt{n!}] + 1)^2$.

Then, Brown's number $([\sqrt{n!}] + 1, n)$ will be a solution of the equation (1). Thus, the condition is sufficient.

Again from equations (2) and (3), we get $(m-1)(m+1) = [\sqrt{n!}]([\sqrt{n!}] + 2) = n! = 2^e \times$ odd number for some $e \in \mathbb{N}$ and $\gcd(m-1, m+1) = d$ where $d = 1$ or 2 since $d \mid (m-1)$ and $d \mid (m+1)$ implies that $d \mid \{(m+1) - (m-1)\}$ i.e. $d \mid 2$. If $d = 1$, then one of $(m-1)$, $(m+1)$ and hence other also will be odd natural number. Thus $n!$ = odd number for $n > 1$, which is a absurd result. Hence $d = 2$ and we can express $n!$ as $n! = 2^k \times$ odd number $= (2a) \times (2^{e-1}b)$ with

a and b relatively odd natural numbers satisfying $\gcd(a, 2^{k-1}b) = 1$ and the difference between $2a$ and $2^{k-1}b$ will be 2.

2.1.1 Note

(i) If we consider the only remaining option for the expression of $n!$ as $n! = ([\sqrt{n!}]-1)([\sqrt{n!}]+1)$, then $[\sqrt{n!}]^2 - 1 = n!$.

Or, $[\sqrt{n!}] = \sqrt{1+n!}$.

$$\text{Thus,} \quad [\sqrt{n!}] = \sqrt{1+n!} = \sqrt{m^2} = m. \tag{4}$$

Now if $\sqrt{n!} = r \in \mathbb{N}$, then $n! = r^2$ and hence $n! + 1 = r^2 + 1 \neq m^2,\ m \in \mathbb{N}$.

Thus $\sqrt{1+n!} = m$ implies that $[\sqrt{n!}] < m$, a contradiction of the equation (4). Hence the expression $n! = ([\sqrt{n!}] - 1)([\sqrt{n!}] + 1)$ can not give any solution of $n! + 1 = m^2;\ n, m \in \mathbb{N}$.

(ii) If $k = m - 1$, then $k^2 < n! = k(k+2) < (k+1)^2$. Thus $[\sqrt{n!}] = k$.

2.1.2 Remark

All the known solutions follow the theorem as shown below.

(i) For $n = 4$, $\sqrt{n!} \approx 4.898979486$. Then $n! + 1 = [\sqrt{n!}]([\sqrt{n!}] + 2) + 1 = (4 \times 6) + 1 = 25 = 5^2 = m^2$.

(ii) For $n = 5$, $\sqrt{n!} \approx 10.95445115$. Then $n! + 1 = [\sqrt{n!}]([\sqrt{n!}] + 2) + 1 = (10 \times 12) + 1 = 121 = 11^2 = m^2$.

(iii) For $n = 7$, $\sqrt{n!} \approx 70.992957397$. Then $n! + 1 = [\sqrt{n!}]([\sqrt{n!}] + 2) + 1 = (70 \times 72) + 1 = 5041 = 71^2 = m^2$.

2.1.3 Corollary

For the solution of equation (1), $m-1 = [\sqrt{n!}]$ should be an even natural number (i.e. m should be odd natural number) k and $k(k+2) = n!$.

Example: (i) if $n = 1$, then $k = [\sqrt{n!}] = 1$ (odd number). Hence no need to check solution of equation (1).

(ii) if $n = 2$, then $k = [\sqrt{n!}] = 1$ (odd number). Hence no need to check solution of equation (1).

(iii) if $n = 3$, then $k = [\sqrt{n!}] = 2$, $k(k+2) = 8 > n!$. Hence equation (1) has no solution for $n = 3$.

(iv) if $n = 4$, then $k = [\sqrt{n!}] = 4$, $k(k+2) = 24 = n!$, $n! + 1 = 5^2$. Hence equation (1) has solution for $n = 4$.

(v) if $n = 5$, then $k = [\sqrt{n!}] = 10$, $k(k+2) = 120 = n!$, $n! + 1 = 11^2$. Hence equation (1) has solution for $n = 5$.

(vi) if $n = 6$, then $k = [\sqrt{n!}] = 26$, $k(k+2) = 728 > n!$. Hence equation (1) has no solution for $n = 6$.

(vii) if $n = 7$, then $k = [\sqrt{n!}] = 70$, $k(k+2) = 5040 = n!$, $n! + 1 = 71^2$. Hence equation (1) has solution for $n = 7$.

(viii) if $n = 8$, then $k = [\sqrt{n!}] = 26$, $k(k+2) = 728 > n!$. Hence equation (1) has no solution for $n = 6$.

(ix) if $n = 9$, then $k = [\sqrt{n!}] = 602$, $k(k+2) = 363608 > n!$. Hence equation (1) has no solution for $n = 9$.

(x) if $n = 10$, then $k = [\sqrt{n!}] = 1904$, $k(k+2) = 3629024 > n!$. Hence equation (1) has no solution for $n = 10$.

(xi) if $n = 11$, then $k = [\sqrt{n!}] = 6317$ (odd number). Hence no need to check solution of equation (1).

From the trend of the above example, we can predict the following result: if n is not a solution of Brocard-Ramanujan problem, then $k(k+2) > n!$ where $k = [\sqrt{n!}]$. It is eventually proved in Theorem 2.4.

2.1.4 Lemma

For a given natural number x, let $x \neq z^2$, for $z \in \mathbb{N}$ and $\sqrt{x} = y + \epsilon$, with $y = [\sqrt{x}]$, $\epsilon = \sqrt{x} - [\sqrt{x}]$ i.e. $0 < \epsilon < 1$. Then $\epsilon(2y + \epsilon)$ is a natural number.

Proof: Given that $\sqrt{x} = y + \epsilon$. Then $x = y^2 + 2y\epsilon + \epsilon^2$. Or, $x - y^2 = 2y\epsilon + \epsilon^2$. Since the left side is a natural number, $\epsilon(2y + \epsilon)$ should also be a natural number.

2.1.5 Remark

We can extend this idea for the n-th ($n \in \mathbb{N}$) root of x. That is if $x \in \mathbb{N}$ with $x \neq z^n$ for $z, n \in \mathbb{N}$ and $x^{\frac{1}{n}} = y + \epsilon$ having $y = [x^{\frac{1}{n}}]$, $\epsilon = x^{\frac{1}{n}} - [x^{\frac{1}{n}}]$ i.e. $0 < \epsilon < 1$, then $x - y^n = \binom{n}{1} y^{n-1}\epsilon + \binom{n}{1} y^{n-2}\epsilon^2 + \cdots + \binom{n}{n}\epsilon^n$. Thus $\binom{n}{1} y^{n-1}\epsilon + \binom{n}{2} y^{n-2}\epsilon^2 + \cdots + \binom{n}{n}\epsilon^n$ is a natural number. For example, if $n = 3$, then $x^{\frac{1}{3}} = y + \epsilon$ with $y = [x^{\frac{1}{3}}]$, $0 < \epsilon < 1$ and $3y^2\epsilon + 3y\epsilon^2 + \epsilon^3$ is a natural number.

2.2 Theorem

In the Brocard-Ramanujan problem, let $\sqrt{n!} = k + \epsilon$ for the natural number n where $\sqrt{n!} \neq m\ (\in \mathbb{N}),\ k = [\sqrt{n!}],\ \epsilon = \sqrt{n!} - [\sqrt{n!}]$ i.e. $0 < \epsilon < 1$. Then $\epsilon(2k + \epsilon)$ must be a natural number.

Proof: Since $\sqrt{n!} \neq m\ (\in \mathbb{N})$, by the Lemma 2.1.4, $n! = k^2 + 2k\epsilon + \epsilon^2$. Or, $n! - k^2 = 2k\epsilon + \epsilon^2$. Thus $\epsilon(2k + \epsilon)$ is a natural number.

For example, (i) if $n = 3$, then $\sqrt{n!} \approx 2.449489743,\ k = 2,\ \epsilon \approx 0.449489743,\ \epsilon(2k + \epsilon) = 2$.

(ii) if $n = 4$, then $\sqrt{n!} \approx 4.898979486,\ k = 4,\ \epsilon \approx 0.898979486,\ \epsilon(2k + \epsilon) = 8$.

2.3 Theorem

Let for the equation (1), $m - 1 = k = [\sqrt{n!}],\ \sqrt{n!} = k + \epsilon$, where $\epsilon = \sqrt{n!} - [\sqrt{n!}]$ i.e. $0 < \epsilon < 1$. Then for the solution of (1), the natural number $\epsilon(2k + \epsilon)$ must satisfy $\epsilon(2k + \epsilon) = 2k$ and conversely. Moreover, $k = \frac{\epsilon^2}{2(1-\epsilon)}$ and ϵ should be distinct and monotonic increasing for all the solutions of (1).

Proof: Since $k = [\sqrt{n!}]$, we know from the Theorem 2.2 that $\epsilon(2k + \epsilon)$ is a natural number. If n is a solution of the equation (1), then $k(k+2) = n!$ by Corollary 2.1.3. Or, $k(k+2) = (k+\epsilon)^2$.

$$\text{Or,} \quad \epsilon(2k + \epsilon) = 2k. \tag{5}$$

Hence the condition is necessary.

Conversely, let $\epsilon(2k + \epsilon) = 2k$. Then $k^2 + 2k\epsilon + \epsilon^2 = k^2 + 2k$.

Or, $n! = k^2 + 2k$ (as $\sqrt{n!} = k + \epsilon$) i.e. $n! + 1 = (k+1)^2$. Thus Brown's number $([\sqrt{n!}] + 1, n)$ will be a solution of the equation (1). Hence the condition is sufficient.

$$\text{From (5), we get} \quad k = \frac{\epsilon^2}{2(1 - \epsilon)} \tag{6}$$

and $\epsilon^2 + 2k\epsilon - 2k = 0$.

Or, $\epsilon = \frac{-2k \pm \sqrt{4k^2+8k}}{2}$.

$$\text{Then,} \quad \epsilon = \sqrt{k^2 + 2k} - k. \tag{7}$$

It is easy to observe that $\epsilon > 0$ and $\epsilon < \sqrt{k^2 + 2k + 1} - k = 1$.

Let $\epsilon_k = f(k) = \sqrt{k^2 + 2k} - k$.

$$\text{Then,} \quad f'(k) = \frac{2k + 2}{2\sqrt{k^2 + 2k}} - 1 > \frac{k + 1}{2\sqrt{k^2 + 2k + 1}} - 1 = 0 \tag{8}$$

Thus $\{f(k)\} = \{\epsilon_k\}$ is strictly monotonic increasing, bounded above as $f(k) < 1 \ \forall \ k$ and also $\lim_{k\to\infty} f(k) = \lim_{k\to\infty} \frac{2k}{\sqrt{k^2+2k}+k} = 1$.

Hence, ϵ should be distinct and monotonic increasing for all the solutions of equation (1).

Let us apply the theorem for the following first few cases.

(i) For $n = 1,\ k = 1,\ \epsilon = 0,\ \frac{\epsilon^2}{2(1-\epsilon)} = 0$ but $k(k+2) \neq n!$. Hence equation (1) has no solution for $n = 1$.

(ii) For $n = 2,\ k = 1,\ \epsilon \approx 0.414213562,\ \frac{\epsilon^2}{2(1-\epsilon)} \approx 0.146446609\ (\notin \mathbb{N}) < k$. Hence equation (1) has no solution for $n = 2$.

(iii) For $n = 3,\ k = 2,\ \epsilon \approx 0.449489743,\ \frac{\epsilon^2}{2(1-\epsilon)} \approx 0.183503419\ (\notin \mathbb{N}) < k$. Hence equation (1) has no solution for $n = 3$.

(iv) For $n = 4,\ k = 4,\ \epsilon \approx 0.898979486,\ \frac{\epsilon^2}{2(1-\epsilon)} = 4$ and $k(k+2) = 24 = n!$. Hence equation (1) has solution for $n = 4$.

(v) For $n = 5,\ k = 10,\ \epsilon \approx 0.95445115,\ \frac{\epsilon^2}{2(1-\epsilon)} = 10$ and $k(k+2) = 120 = n!$. Hence equation (1) has solution for $n = 5$.

(vi) For $n = 6,\ k = 26,\ \epsilon \approx 0.83281573,\ \frac{\epsilon^2}{2(1-\epsilon)} \approx 2.07430412\ (\notin \mathbb{N}) < k$ and also $k(k+2) \neq n!$. Hence equation (1) has no solution for $n = 6$.

(vii) For $n = 7,\ k = 70,\ \epsilon \approx 0.9929573972,\ \frac{\epsilon^2}{2(1-\epsilon)} = 70$ and $k(k+2) = 5040 = n!$. Hence equation (1) has solution for $n = 7$.

(viii) For $n = 8,\ k = 200,\ \epsilon \approx 0.7984064,\ \frac{\epsilon^2}{2(1-\epsilon)} \approx 1.581033892\ (\notin \mathbb{N}) < k$ and also $k(k+2) \neq n!$. Hence equation (1) has no solution for $n = 8$.

(ix) For $n = 9,\ k = 602,\ \epsilon \approx 0.3952191,\ \frac{\epsilon^2}{2(1-\epsilon)} \approx 0.1291361398\ (\notin \mathbb{N}) < k$ and also $k(k+2) \neq n!$. Hence equation (1) has no solution for $n = 9$.

(x) For $n = 10,\ k = 1904,\ \epsilon \approx 0.940944,\ \frac{\epsilon^2}{2(1-\epsilon)} \approx 7.496063447\ (\notin \mathbb{N}) < k$ and also $k(k+2) \neq n!$. Hence equation (1) has no solution for $n = 10$.

(xi) For $n = 11,\ k = 6371,\ \epsilon \approx 0.974359,\ \frac{\epsilon^2}{2(1-\epsilon)} \approx 18.51278095\ (\notin \mathbb{N}) < k$ and also $k(k+2) \neq n!$. Hence equation (1) has no solution for $n = 11$.

2.3.1 Remark

(i) The function $f(k)$ has no optimum value in $0 < \epsilon < 1$. For the known solutions of Brocard-Ramanujan problem (c.f. equation (1)), (i) $\epsilon \approx 0.898979486$ for $n = 3$ (ii) $\epsilon \approx 0.95445115$ for $n = 5$ (iii) $\epsilon \approx 0.9929573972$ for $n = 7$. The next value of ϵ should be more than $\epsilon \approx 0.9929573972$ as ϵ should be distinct and monotonic increasing for all the solutions of equation

(1). However, ϵ is neither monotonic increasing nor monotonic decreasing for all natural numbers n (c.f. examples of Theorem 2.3). Also, ϵ is a function of k, $k = [\sqrt{n!}]$ in the form of the equation (7) and according to Bruce Berndt and William Galway [10], there is no new solution of the problem based on computations up to $n = 10^9$. If we consider a new solution of Brocard-Ramanujan problem as $n \geq 10^5$, then value of ϵ should be more than $\epsilon \approx 0.999\cdots 905915$ (**digit 0 is coming after 228287 numbers of 9 digit, which takes more than 66 pages in LibreOffice Writer**) for $k = [\sqrt{10^5!}]$ according to the calculation in MATHEMATICA 8. If we consider $n \geq 10^9$ as new solution of Brocard-Ramanujan problem, then it is obvious that the numbers of 9 digit in the form $\epsilon = 0.999\cdots$ will be much higher than the corresponding case of $n \geq 10^5$, but I am unable to calculate the numbers of 9 digit in the value of ϵ in my personal laptop (with 8GB Ram) using MATHEMATICA 8. Hence, it suggests that there will be no new solution of the problem.

(ii) For the non-solution of (1), what will be the expression of the natural number $\epsilon(2k+\epsilon)$?. The said expression is provided in theorem 2.13.

2.3.2 Corollary

For the solution of Brocard-Ramanujan problem, we have $k = \frac{\epsilon^2}{2(1-\epsilon)}$. Hence, $\{g(\epsilon)\} = \{k_\epsilon\}$ is also monotonic increasing.

2.4 Theorem

For any natural number n, $n! \leq k(k+2)$, where $k = [\sqrt{n!}]$ and if n is not a solution of Brocard-Ramanujan problem (c.f. equation (1)), then $n! < k(k+2)$.

Proof: For any natural number n, $\sqrt{n!} = k + \epsilon$ with $k = [\sqrt{n!}]$ and $0 \leq \epsilon < 1$. Then $n! = k^2 + 2k\epsilon + \epsilon^2 < k^2 + 2k + 1$ as $\epsilon < 1$. We know that $n!$, k^2 and $2k\epsilon + \epsilon^2$ (by Theorem 2.3) are integers. Hence $n! \leq k(k+2)$, $\forall\ n \in \mathbb{N}$.

We also know from Corollary 2.1.3 that for the solution of Brocard-Ramanujan problem, we have $n! = k(k+2)$. Thus if n is not a soluton of Brocard-Ramanujan problem (c.f. equation (1)), then $n! < k(k+2)$.

2.4.1 Corollary

If natural number n is not a solution of Brocard-Ramanujan problem (c.f. equation (1)), then $2k\epsilon + \epsilon^2 < 2k$.

2.5 Theorem

The Brocard-Ramanujan problem (c.f. equation (1)) has only finitely many solutions.

Proof: From Theorem 2.3, we know that for the solution of (1), the natural number $2k\epsilon+\epsilon^2$ must satisfy $2k\epsilon + \epsilon^2 = 2k$. Our endeavor should be to find all possible k and corresponding ϵ such that $2k\epsilon + \epsilon^2 = 2k$ for the Brocard-Ramanujan problem. Moreover, if n is not a solution of Brocard-Ramanujan problem, then $2k\epsilon + \epsilon^2 < 2k$ by Corollary 2.4.1.

Let $f(\epsilon, k) = 2k\epsilon + \epsilon^2$. Then by Taylor's expansion of two variables with respect to (ϵ_0, k_0), we obtain $f(\epsilon, k) = f(\epsilon_0, k_0) + (\epsilon - \epsilon_0)\frac{\partial f}{\partial \epsilon}(\epsilon_1, k_1) + (k - k_0)\frac{\partial f}{\partial k}(\epsilon_1, k_1) = 2k_0\epsilon_0 + \epsilon_0^2 + (\epsilon - \epsilon_0)(2\epsilon_1 + 2k_1) + (k - k_0)(2\epsilon_1)$ with $k_0 = [\sqrt{7!}] = 70,\ \epsilon_0 = \sqrt{7!} - [\sqrt{7!}] \approx 0.9929573972$ and for some ϵ_1 and k_1 satisfying $\epsilon_0 < \epsilon_1 < \epsilon < 1,\ k_0 < k_1 < k$.

Thus $f(\epsilon, k) = 140 + (\epsilon - \epsilon_0)(2\epsilon_1 + 2k_1) + 2k + 2k(\epsilon_1 - 1) - 140\epsilon_1 = 2k + 140(1 - \epsilon_1) + (\epsilon - \epsilon_0)(2\epsilon_1 + 2k_1) - 2k(1 - \epsilon_1) < 2k$ if $140(1 - \epsilon_1) + (\epsilon - \epsilon_0)(2\epsilon_1 + 2k_1) - 2k(1 - \epsilon_1) < 0$ i.e. $k > \frac{140(1-\epsilon_1)+2(\epsilon-\epsilon_0)(\epsilon_1+k_1)}{2(1-\epsilon_1)} = 70 + \frac{(\epsilon-\epsilon_0)(\epsilon_1+k_1)}{(1-\epsilon_1)}$.

Thus $f(\epsilon, k) < 2k$ for all $k > 70 + \frac{(1-\epsilon_0)(\epsilon_1+k_1)}{(1-\epsilon_1)}$.

Then, the Brocard-Ramanujan problem (c.f. equation (1)) has no solutions if $k > 70 + \frac{(1-\epsilon_0)(\epsilon_1+k_1)}{(1-\epsilon_1)}$.

Hence, the Brocard-Ramanujan problem has only finitely many solutions.

2.6 Theorem

Let for the equation (1), $m - 1 = k = [\sqrt{n!}]$, $\sqrt{n!} = k + \epsilon$, where $\epsilon = \sqrt{n!} - [\sqrt{n!}]$ i.e. $0 < \epsilon < 1$. Then for the solution of (1), $n! = \frac{\epsilon^2(2-\epsilon)^2}{4(1-\epsilon)^2}$ i.e. the solutions $(m, n!)$ will be in the form $(\frac{\epsilon^2}{2(1-\epsilon)} + 1, \frac{\epsilon^2(2-\epsilon)^2}{4(1-\epsilon)^2})$. Moreover, for all the solutions of (1), $x = n!$ and $y = m - 1 = k = [\sqrt{n!}]$ should satisfy

$$y^4 + 4y^3 + 2xy^2 + 4y^2 + 4xy - 3x^2 = 0,\ y^3 + 3y^2 + 2y - xy - x = 0. \tag{9}$$

Proof: For the solution of Brocard-Ramanujan problem, we have $k = \frac{\epsilon^2}{2(1-\epsilon)}$. Then $\sqrt{n!} = k + \epsilon = \frac{\epsilon(2-\epsilon)}{2(1-\epsilon)}$.

$$\text{Thus, } n! = \frac{\epsilon^2(2-\epsilon)^2}{4(1-\epsilon)^2}. \tag{10}$$

Or, $4n!(1 - 2\epsilon + \epsilon^2) = \epsilon^2(4 - 4\epsilon + \epsilon^2)$.

Or, $4n!\{1-2(\sqrt{n!}-[\sqrt{n!}])+(\sqrt{n!}-[\sqrt{n!}])^2\} = \{(\sqrt{n!}-[\sqrt{n!}])^2(4-4(\sqrt{n!}-[\sqrt{n!}])+(\sqrt{n!}-[\sqrt{n!}])^2\}$, where $\epsilon = \sqrt{n!} - [\sqrt{n!}]$.

Or, $4n!\{(1+2[\sqrt{n!}]+n!+[\sqrt{n!}]^2\}-8n!\sqrt{n!}\{1+[\sqrt{n!}]\}=\{(4n!+4[\sqrt{n!}]^2+12n![\sqrt{n!}]+n!^2+6n![\sqrt{n!}]^2+4[\sqrt{n!}]^3+[\sqrt{n!}]^4)-4n!\sqrt{n!}(2[\sqrt{n!}]+n!+3[\sqrt{n!}]^2+n![\sqrt{n!}]+[\sqrt{n!}]^3)\}$.

Comparing both sides, we get $4n!\{(1+2[\sqrt{n!}]+n!+[\sqrt{n!}]^2\}=\{(4n!+4[\sqrt{n!}]^2+12n![\sqrt{n!}]+n!^2+6n![\sqrt{n!}]^2+4[\sqrt{n!}]^3+[\sqrt{n!}]^4),\ 2n!+2n![\sqrt{n!}]=2[\sqrt{n!}]+n!+3[\sqrt{n!}]^2+n![\sqrt{n!}]+[\sqrt{n!}]^3$.

$$\text{Thus, } y^4+4y^3+2xy^2+4y^2+4xy-3x^2=0, \tag{11}$$

$$y^3+3y^2+2y-xy-x=0; \tag{12}$$

where $x=n!,\ y=[\sqrt{n!}]$.

All the known solutions of Brocard-Ramanujan problem satisfy the equations (11) and (12) as shown below.

(i) For $n=4,\ x=24,\ y=4$. Then $y^4+4y^3+2xy^2+4y^2+4xy-3x^2=0$ and $y^3+3y^2+2y-xy-x=0$.

(ii) For $n=5,\ x=120,\ y=10$. Then $y^4+4y^3+2xy^2+4y^2+4xy-3x^2=0$ and $y^3+3y^2+2y-xy-x=0$.

(iii) For $n=7,\ x=5040,\ y=70$. Then $y^4+4y^3+2xy^2+4y^2+4xy-3x^2=0$ and $y^3+3y^2+2y-xy-x=0$.

2.6.1 Remark

It is shown that all solutions of Brocard-Ramanujan problem satisfy the equations (11) and (12). By Bézout theorem, (c.f. Sturmfels [17]) for a general square polynomial system $f_1(x_1,x_2,\cdots,x_n)=0,\ f_2(x_1,x_2,\cdots,x_n)=0,\ \cdots,\ f_n(x_1,x_2,\cdots,x_n)=0$ having $d_1,d_2,\cdots,d_n$ as the degrees of $f_1,f_2,\cdots,f_n$ respectively; the number of its isolated zeros in $\mathbb{C}^n$, counting multiplicities, does not exceed the number $d=d_1d_2\cdots d_n$ except the system has an infinite number of zeros.

However, in-built functions 'Solve', 'Solve with integer', 'NSolve', 'NSolve with real', 'NSolve with integer', 'FindRoot' are unable to provide any solutions of Brocard-Ramanujan problem. To solve by hands, the following approaches are considered.

Approach 1: If we solve equation (12), then we get $y=-1,\ x=y(y+2)$. For $y=-1$, we get from equation (11) as $3x^2+2x-1=0$ i.e. $x=-1,\frac{1}{3}$.

For $x=y(y+2)$, we get from equation (11) as $(4y+2y^2)y(y+2)-3y^2(y+2)^2+4y^2+4y^3+y^4=0$. Or, $2y^2(y+2)^2-3y^2(y+2)^2+y^2(y+2)^2=0$ Or, $0=0$, which gives no explicit solution in this case. However, we can get three known solutions of the

Brocard-Ramanujan problem as $(24, 4),\ (120, 10),\ (5040, 70)$ and infinite number of integer solutions in the form (c, d) with $d < 0,\ d \in \mathbb{Z},\ c = d^2 - 2d$. First ten of these solutions are $(-1, -1),\ (0, -2),\ (3, -3),\ (8, -4),\ (15, -5),\ (24, -6),\ (35, -7),\ (48, -8),\ (63, -9),\ (80, -10)$.

Approach 2: From equation (11), we have $3x^2 - 2x(y^2 + 2y) - (4y^2 + 4y^3 + y^4) = 0$. Or, $x = \frac{2y(y+2) \pm \sqrt{4y^2(y+2)^2 + 12(4y^2+4y^3+y^4)}}{6} = \frac{y(y+2) \pm 2y(y+2)}{3} = y(y+2),\ -\frac{y(y+2)}{3}$.

If $x = y(y + 2)$, then the equation (12) gives $-(y + 1)y(y + 2) + y^3 + 3y^2 + 2y = 0$. Or, $-y(y + 1)y(y + 2) + y(y + 1)y(y + 2) = 0$. Or, $0 = 0$, which gives no explicit solution in this case. But, we can find solutions as $(24, 4),\ (120, 10),\ (5040, 70)$ and (c, d) with $d < 0,\ d \in \mathbb{Z}$, $c = d^2 - 2d$.

If $x = -\frac{y(y+2)}{3}$, then the equation (12) gives $-\frac{(y+1)y(y+2)}{3} + y^3 + 3y^2 + 2y = 0$. Or, $-\frac{(y+1)y(y+2)}{3} + (y + 1)y(y + 2) = 0$. Or, $\frac{2(y+1)y(y+2)}{3} = 0$. Or, $y = 0, -1, -2$. Then $x = 0, \frac{1}{3}, 0$ respectively.

Approach 3: From equation (11), we have $y^4 + 4y^3 + (2x + 4)y^2 + 4xy - 3x^2 = 0$.

$$\text{Or, } (y^2 + 2y + \lambda)^2 - (my + n)^2 = 0 \tag{13}$$

with $4 + 2\lambda - m^2 = 2x + 4,\ 4\lambda - 2mn = 4x,\ \lambda^2 - n^2 = -3x^2$ i.e. $m^2 = 2(\lambda - x),\ mn = 2(\lambda - x),\ n^2 = \lambda^2 + 3x^2$. Then $m^2 = mn$ i.e. $m = 0, n$.

For $m = 0,\ \lambda = x,\ n^2 = 4x^2$ i.e. $n = \pm 2x$. Then from equation (13), we have $(y^2 + 2y + x)^2 - (\pm 2x)^2 = 0$. Or, $(y^2 + 2y + 3x)(y^2 + 2y - x) = 0$. Or, $x = y(y + 2),\ -\frac{y(y+2)}{3}$. If $x = y(y + 2)$, then the equation (12) gives $-(y + 1)y(y + 2) + y^3 + 3y^2 + 2y = 0$. Or, $-y(y + 1)y(y + 2) + y(y + 1)y(y + 2) = 0$. Or, $0 = 0$, which gives no explicit solution in this case and we can find solutions as $(24, 4),\ (120, 10),\ (5040, 70)$ and (c, d) with $d < 0,\ d \in \mathbb{Z}$, $c = d^2 - 2d$.

If $x = -\frac{y(y+2)}{3}$, then the equation (12) gives $-\frac{(y+1)y(y+2)}{3} + y^3 + 3y^2 + 2y = 0$. Or, $-\frac{(y+1)y(y+2)}{3} + y(y + 1)y(y + 2) = 0$. Or, $\frac{2(y+1)y(y+2)}{3} = 0$. Or, $y = 0, -1, -2$.

For $m = n$, $n^2 = 2(\lambda - x)$ and $n^2 = \lambda^2 + 3x^2$ i.e. $2(\lambda - x) = \lambda^2 + 3x^2$. Or, $\lambda^2 - 2\lambda + 2x + 3x^2 = 0$. Or, $\lambda = \frac{2 \pm \sqrt{4 - 4(3x^2+2x)}}{2} = \frac{1 \pm \sqrt{1 - (3x^2+2x)}}{2}$. We can see that $1 - (3x^2 + 2x) < 0$ for $x > 1$.

Hence there are solutions $(0, 0),\ (\frac{1}{3}, -1),\ (24, 4),\ (120, 10),\ (5040, 70)$ and (c, d) with $d < 0,\ d \in \mathbb{Z}, c = d^2 - 2d$ of equations (11) and (12); among them only three solutions $(24, 4),\ (120, 10),\ (5040, 70)$ are of the Brocard-Ramanujan problem which are already known.

2.7 Theorem

For the fourth solution of the Brocard-Ramanujan problem (1), $m = \alpha \times 10^{\beta} \pm 1$ for some $\alpha,\ \beta \in \mathbb{N}$ (i) with $\beta \geq 24999$ when $n \geq 10^5$ and (ii) with $\beta \geq 249999998$ when $n \geq 10^9$.

Proof: We know that $n! = (m-1)(m+1)$ with gcd $d = (m-1, m+1) = 2$ and $2_1^{\beta}/n!$ but 2^{β_1+1} not divide $n!$ where $\beta_1 = [\frac{n}{2}] + [\frac{n}{4}] + [\frac{n}{8}] + \cdots$. Then $2^{\beta_1-1}/(m-1)$ or $2^{\beta_1-1}/(m+1)$ but not both. Again, $5^{\beta_3}/n!$ but 5^{β_3+1} not divide $n!$ where $\beta_3 = [\frac{n}{5}] + [\frac{n}{25}] + [\frac{n}{125}] + \cdots$. and $5^{\beta_3}/(m-1)$ or $5^{\beta_3}/(m+1)$ but not both.

If $2/(m-1),\ 5^{\beta_3}/(m-1)$ and $2^{\beta_1-1}/(m+1)$, then $10/(m-1)$ but 100 not divide $(m-1)$ and 10 does not divide $m+1$ which contradicts $100/n!$ for $n \geq 10$ as $d = 2$. Also for $2/(m+1),\ 5^{\beta_3}/(m+1)$ and $2^{\beta_1-1}/(m-1)$, then $10/(m+1)$ but 100 not divide $(m+1)$ and 10 does not divide $m-1$ which contradicts $100/n!$ for $n \geq 10$ as $d = 2$. Hence if $2^{\beta_1-1}/(m-1)$, them $5^{\beta_3}/(m-1)$ and if $2^{\beta_1-1}/(m+1)$, them $5^{\beta_3}/(m+1)$.

(i) For $n = 10^5$, we can calculate as $\beta_3 = 24999$ and $\beta_1 > \beta_3$. Thus m will be in the form $m = \alpha \times 10^{\beta} \pm 1$ for some $\alpha,\ \beta \in \mathbb{N}$ with $\beta \geq 24999$ when $n \geq 10^5$.

(ii) For $n = 10^9$, we can evaluate as $\beta_3 = 249999998$ and $\beta_1 > \beta_3$. Thus m will be in the form $m = \alpha \times 10^{\beta} \pm 1$ for some $\alpha,\ \beta \in \mathbb{N}$ with $\beta \geq 249999998$ when $n \geq 10^9$.

2.8 Theorem

For the fourth solution of the Brocard-Ramanujan problem (1), $m = 2^{\beta_1-1}5^{\beta_3}m_3 + 1$ or $m = 2^{\beta_1-1}5^{\beta_3}m_3 - 1$, m_3 is odd.

Proof: For the Brocard-Ramanujan problem (1), we have $n! = m^2 - 1 = (m-1)(m+1)$, where m is odd.

Case I: $m = 4m_1 + 1$. Then $n! = 4 \times 2m_1 \times (2m_1 + 1)$.

Case I(i): $m_1 = 2m_2$. Then $m = 8m_2 + 1,\ n! = 4 \times 4m_2 \times (4m_2 + 1) = 4 \times 2^{\beta_1-2}5^{\beta_3}m_3 \times (2^{\beta_1-2}5^{\beta_3}m_3 + 1)$ using theorem (2.7), where $m_2 = 2^{\beta_1-4}5^{\beta_3}m_3,\ m = 2^{\beta_1-1}5^{\beta_3}m_3 + 1$ and m_3 is odd.

Case I(ii): $m_1 = 2m_2 + 1$. Then $m = 8m_2 + 5,\ n! = 4 \times 2(2m_2 + 1) \times (4m_2 + 3)$. Then here $\beta_1 = 3$ and so no fourth solution possible in this case as $\beta_1 \geq 999999987$ when $n \geq 10^9$.

Case II: $m = 4m_1 + 3$. Then $n! = 4 \times (2m_1 + 1) \times 2(m_1 + 1)$.

Case II(i): $m_1 = 2m_2$. Then $m = 8m_2 + 5,\ n! = 4 \times (4m_2 + 1) \times 2(2m_2 + 1)$. Then here $\beta_1 = 3$ and so no fourth solution possible in this case as $\beta_1 \geq 999999987$ when $n \geq 10^9$.

Case II(ii): $m_1 = 2m_2 + 1$. Then $m = 8m_2 + 7,\ n! = 4 \times (4m_2 + 3) \times 4(m_2 + 1) =$

$4 \times (2^{\beta_1-2}5^{\beta_3}m_3 - 1) \times 2^{\beta_1-2}5^{\beta_3}m_3$ using theorem (2.7), where $m_2 + 1 = 2^{\beta_1-4}5^{\beta_3}m_3,\ m = 2^{\beta_1-1}5^{\beta_3}m_3 - 1$ and m_3 is odd.

2.9 Theorem

For all the solutions of the Brocard-Ramanujan problem (1), m should be in the form of $m = 5+6t$ and $m = 11+30t$ for all the solutions of the Brocard-Ramanujan problem (1) except first, while $m = 71 + 5040t$ for all the solutions starting from third.

Proof: From the result [20], we know that if $d_1,\ d_2 \in \mathbb{N}$ and $e_1,\ e_2 \in \mathbb{Z}$; then $y = e_1 \pmod{d_1}$, $y = e_2 \pmod{d_2}$ have a simultaneous solution iff gcd $(d_1, d_2)/(e_1 - e_2)$ and if this condition be satisfied; the solution is unique module lcm (d_1, d_2).

For the solution of the Brocard-Ramanujan problem (1), we have $n! = m^2 - 1$. Then $m = 1 \pmod{\frac{k}{2}}$ and $m = -1 \pmod{\frac{k+2}{2}}$ with gcd $(\frac{k}{2}, \frac{k+2}{2}) = 1$.

Case I: For the Brown's number $(5, 4)$, $m = 5,\ k = 4$, $m = 1 \pmod 2$ and $m = -1 \pmod 3$ and the solution is $f_t = m + 6t = 5 + 6t$. When $t = 1,\ g_1 = 11$ and also $11 = 1 \pmod 2$, $11 = -1 \pmod 3$ giving m for 2nd solution of (1). For $t = 1,\ g_1 1 = 71$ and also $71 = 1 \pmod 2$, $71 = -1 \pmod 3$ which provides m for 3rd solution of (1).

Case II: For the Brown's number $(11, 5)$, $m = 11,\ k = 10$, $m = 1 \pmod 5$ and $m = -1 \pmod 6$ and the solution is $f_t = m + 30t = 11 + 30t$. When $t = 2,\ g_2 = 71$ and also $71 = 1 \pmod 5$, $71 = -1 \pmod 6$ giving m for 3rd solution of (1).

Case III: For the Brown's number $(71, 7)$, $m = 71,\ k = 70$, $m = 1 \pmod{35}$ and $m = -1 \pmod{36}$ and the solution is $f_t = m + 5040t = 71 + 5040t$.

2.9.1 Note

In all three cases, n can be evaluated as $n! = m^2 - 1$.

2.10 Theorem

If $\sqrt{n!} = k + \epsilon$ for solution of Brocard-Ramanujan problem (c.f. equation (1)), then necessary condition is k divides $n!$, while for sufficient condition $\frac{n!}{k} = k+2$. If $2k\epsilon+\epsilon^2 = k$, then there will be no other solution of (1) in this case. But computation suggests that $n = 3$ may be only one example for which $2k\epsilon + \epsilon^2 = k$. If $2k\epsilon + \epsilon^2 = k$, then $\epsilon < 0.5$ and $\epsilon,\ k$ are strictly monotonic increasing for the solutions of $n! = k(k + 1)$.

Proof: $n! = k(k+1)$ If n is a solution of the equation (1), then $k(k+2) = n!$ by Corollary 2.1.3. Thus necessary condition of (1) is that k divides $n!$. But it is not sufficient condition as we need equation $\frac{n!}{k} = k+2$. Then $n!+1 = (k+1)^2$. Hence the sufficient condition is $\frac{n!}{k} = k+2$.

If $2k\epsilon + \epsilon^2 = k$, then $n! = k^2 + 2k\epsilon + \epsilon^2 = k(k+1)$ and $\frac{n!}{k} = k+1$. Then there will be no other solution of (1) in this case.

If $2k\epsilon + \epsilon^2 = k$, it is clear that $\epsilon < 0.5$ and $\epsilon \longrightarrow 0.5$ as $n \longrightarrow \infty$. Also $\epsilon = \sqrt{k^2+k} - k$. Or, $\frac{d\epsilon}{dk} = \frac{2k+1}{2\sqrt{k^2+k}} - 1$ and $\left(\frac{2k+1}{2\sqrt{k^2+k}}\right)^2 = \frac{4k^2+4k+1}{4k^2+4k} > 1$. Then $\frac{d\epsilon}{dk} > 0$. Hence ϵ is strictly monotonic increasing for the solutions of $n! = k(k+1)$. Again, $k = \frac{\epsilon^2}{1-2\epsilon}$, $\frac{dk}{d\epsilon} = \frac{2\epsilon(1-\epsilon)}{(1-2\epsilon)^2} > 0$. So, k is strictly monotonic increasing for the solutions of $n! = k(k+1)$

We know that for $n = 3$, $3! = 6 = 2(2+1)$, $k = [\sqrt{3!}] = 2$, $\epsilon \approx 0.4494897428$, $2k\epsilon + \epsilon^2 = 2 = k$.

If $n = 17$, $k = 18859677$. when ϵ satisfies $2k\epsilon + \epsilon^2 = k$, then $\epsilon \approx 0.499999993$. But actual value of ϵ for $n = 17$ is $\epsilon \approx 0.306253$.

2.11 Theorem

If $\sqrt{n!} = k + \epsilon$ for solution of Brocard-Ramanujan problem (c.f. equation (1)), then (ϵ, k) lies on hyperbola.

Proof: By the Theorem 2.3 for the solution of (1), the natural number $\epsilon(2k+\epsilon)$ must satisfy $\epsilon(2k+\epsilon) = 2k$ i.e. $\epsilon^2 + 2\epsilon k - 2k = 0$, which is a hyperbola with center $(1, -1)$. Let us consider the translation $\epsilon = \epsilon_1 + 1$, $k = k_1 - 1$. Then the hyperbola is shifted to $\epsilon_1^2 + 2\epsilon_1 k_1 + 1 = 0$. Using the rotation $\epsilon_1 = \epsilon_2 \cos\theta - k_2 \sin\theta$, $k_1 = \epsilon_2 \sin\theta + k_2 \cos\theta$, we get the transformed hyperbola as $(\cos^2\theta + 2\sin\theta\cos\theta)\epsilon_2^2 + (\sin^2\theta - 2\sin\theta\cos\theta)k_2^2 + (-2\sin\theta\cos\theta + 2\cos^2\theta - 2\sin^2\theta)\epsilon_2 k_2 + 1 = 0$. For the value of θ, we consider coefficient of $\epsilon_2 k_2$ as zero. Then $-2\sin\theta\cos\theta + 2\cos^2\theta - 2\sin^2\theta = 0$. Or, $\tan^2\theta + \tan\theta - 1 = 0$. Or, $\tan\theta = \frac{-1\pm\sqrt{5}}{2}$.

For $\tan\theta = \frac{\sqrt{5}-1}{2}$, $\sin\theta = \frac{\sqrt{5}-1}{\sqrt{10-2\sqrt{5}}}$, $\cos\theta = \frac{2}{\sqrt{10-2\sqrt{5}}}$. Then the transformed hyperbola is $\frac{3-\sqrt{5}}{\sqrt{5}-1}k_2^2 - \frac{2}{\sqrt{5}-1}\epsilon_2^2 = 1$, where $\epsilon = 1 + \frac{2}{\sqrt{10-2\sqrt{5}}}\epsilon_2 - \frac{\sqrt{5}-1}{\sqrt{10-2\sqrt{5}}}k_2$ and $k = -1 + \frac{\sqrt{5}-1}{\sqrt{10-2\sqrt{5}}}\epsilon_2 + \frac{2}{\sqrt{10-2\sqrt{5}}}k_2$.

For $\tan\theta = \frac{-\sqrt{5}-1}{2}$, $\sin\theta = \frac{\sqrt{5}+1}{\sqrt{10+2\sqrt{5}}}$, $\cos\theta = -\frac{2}{\sqrt{10+2\sqrt{5}}}$ or $\sin\theta = -\frac{\sqrt{5}+1}{\sqrt{10+2\sqrt{5}}}$, $\cos\theta = \frac{2}{\sqrt{10+2\sqrt{5}}}$. Then the transformed hyperbola is $\frac{2}{\sqrt{5}+1}\epsilon_2^2 - \frac{3+\sqrt{5}}{\sqrt{5}+1}k_2^2 = 1$, where $\epsilon = 1 - \frac{2}{\sqrt{10+2\sqrt{5}}}\epsilon_2 - \frac{\sqrt{5}+1}{\sqrt{10+2\sqrt{5}}}k_2$ and $k = -1 + \frac{\sqrt{5}+1}{\sqrt{10+2\sqrt{5}}}\epsilon_2 - \frac{2}{\sqrt{10+2\sqrt{5}}}k_2$ or $\epsilon = 1 + \frac{2}{\sqrt{10+2\sqrt{5}}}\epsilon_2 + \frac{\sqrt{5}+1}{\sqrt{10+2\sqrt{5}}}k_2$ and $k = -1 - \frac{\sqrt{5}+1}{\sqrt{10+2\sqrt{5}}}\epsilon_2 + \frac{2}{\sqrt{10+2\sqrt{5}}}k_2$ respectively.

2.11.1 Note

(i) $2k = \epsilon^2 + 2k\epsilon < 3k\epsilon$ i.e. $\epsilon > \frac{2}{3}$ for the solution of (1).

(ii) $2k = \epsilon^2 + 2k\epsilon < \frac{k\epsilon}{2} + 2k\epsilon$ as $k \geq 2$ for $n \geq 3$ i.e. $\epsilon > \frac{4}{5}$ for the solution of (1).

(iii) $2k = \epsilon^2 + 2k\epsilon < 1 + 2k\epsilon$ i.e. $\epsilon > \frac{2k-1}{2k}$. For $n = 4$, $k = 4$ and hence $\epsilon > \frac{7}{8}$ for the solution of (1) if $n \geq 4$.

(iv) $2k = \epsilon^2 + 2k\epsilon < \epsilon + 2k\epsilon$ i.e. $\epsilon > \frac{2k}{2k+1}$. For $n = 4$, $k = 4$ and hence $\epsilon > \frac{8}{9}$ for the solution of (1) if $n \geq 4$. It can be noted that $\frac{2k}{2k+1}$ better approximation of ϵ than $\frac{2k-1}{2k}$.

(v) $\epsilon^2 + 2k\epsilon \geq (1 + 2k)\left(\epsilon^{2k}\epsilon^2\right)^{\frac{1}{2k+1}}$. Or, $(2k+1)\epsilon^{\frac{2k+2}{2k+1}} \leq \epsilon^2 + 2k\epsilon < g(2k+1)$ for some $0 < g < 1$. Then $\epsilon < g^{\frac{2k+1}{2k+2}}$. Also $2k = \epsilon^2 + 2k\epsilon < g(2k+1)$ i.e. $g > \frac{2k}{2k+1}$.

2.12 Theorem

If $n! = a^2b = a^2(c^2 + d),\ d \leq 2c,\ n \in \mathbb{N}$ and $n! + 1 = (ac + e)^2$, then for the solution of Brocard-Ramanujan problem (c.f. (1)), $1 \leq d < 1 + \frac{2ce}{a}$.

Proof: For natural number $n(> 1)$, we know that

$$n! = a^2b = a^2(c^2 + d),\ d \leq 2c. \tag{14}$$

$$\text{Then } n! + 1 = a^2(c^2 + d) + 1 = (ac + e)^2 \tag{15}$$

$$\text{if } a^2d + 1 = 2ace + e^2. \tag{16}$$

$$\text{For } e = 1, ad = 2c. \tag{17}$$

$$\text{If } e > 1, a(ad - 2ce) = e^2 - 1. \tag{18}$$

$$\text{Thus } a/(e^2 - 1) \text{ and } ad - 2ce > 0 \text{ i.e. } e < \frac{ad}{2c} \tag{19}$$

$$\text{Or, } e < a \text{ as } d \leq 2c. \tag{20}$$

We observe that $ad - 2ce \geq a$ or $ad - 2ce < a$.

$$\text{If } ad - 2ce \geq a \textit{ i.e. } a(d-1) \geq 2ce, \tag{21}$$

$$\text{then } d > 1 \text{ as } ce > 0 \text{ and } d > 1 + \frac{2ce}{a} \tag{22}$$

But then in (18), L.H.S.$> a^2$ and R.H.S.$< a^2$. Thus (16) and hence (15) has no solution.

$$\text{If } ad - 2ce < a,\ \textit{then } a(d-1) < 2ce, \tag{23}$$

When $d < 1$ (i.e. $d = 0$), then (18) is always true, but $d = 0$ contradicts (19) as $ce > 0$. Also $n! \neq a^2c^2$ if $n > 1$.

If $d = 1$, $ce > 0$ from (23). Using (19) and (23), we get $0 < a-2ce < a$. Thus (16) and hence (15) has solution if it satisfies (18). From (18), we have $1 < d < 1 + \frac{2ce}{a}$ and $0 < ad - 2ce < a$. This means (16) and hence (15) has solution if it satisfies (18).

2.12.1 Remark

For $n \geq 6$; $a^2 = a_1^2 n^2$ if n is not prime number. $a^2 = a_1^2(n+1)^2$ if n is a prime number ($n \neq 7$) but $\frac{n+1}{2}$ is not a prime number and $a^2 = a_1^2(n+2)^2$ if n, $\frac{n+1}{2}$ are both prime numbers following the examples. If $n = 7$, then $a^2 = a_1^2(n+5)^2 = 1^2 12^2$. Moreover, $\sqrt{n!} = a(c + \epsilon_3)$, $2c\epsilon_3 + \epsilon_3^2 = d$, $(0 < \epsilon_3 < 1)$. Thus satisfying the Theorem 2.3, we have $k = ac + [a\epsilon_3]$, $\epsilon = a\epsilon_3 - [a\epsilon_3]$.

1. $1! = 1 = 1^2$, $a = 1$, $b = 1$, $c = 1$, $d = 0$, $\epsilon_3 = 0$, $k = 1$, $\epsilon = 0$.

2. $2! = 2 = 1^2(1^2 + 1)$, $a = 1$, $b = 1$, $c = 1$, $d = 1$, $\epsilon_3 \approx 0.414213562$, $k = 1$, $\epsilon \approx 0.414213562$.

3. $3! = 2 \times 3 = 1^2(2^2 + 2)$, $a = 1$, $b = 6$, $c = 2$, $d = 2$, $\epsilon_3 \approx 0.449489743$, $k = 2$, $\epsilon \approx 0.449489743$.

4. $4! = 2^2 \times 2 \times 3 = 2^2(2^2 + 2)$, $a = 2$, $b = 6$, $c = 2$, $d = 2$, $\epsilon_3 \approx 0.449489743$, $k = 4$, $\epsilon \approx 0.898979486$.

5. $5! = 2^2 \times 2 \times 3 \times 5 = 2^2(5^2 + 5)$, $a = 2$, $b = 30$, $c = 5$, $d = 5$, $\epsilon_3 \approx 0.477225575$, $k = 10$, $\epsilon \approx 0.95445115$.

6. $6! = 4^2 3^2 5 = 2^2 6^2 5 = 12^2(2^2 + 1)$, $a = 12$, $a_1 = 2$, $b = 5$, $c = 2$, $d = 1$, $\epsilon_3 \approx 0.236067977$, $k = 26$, $\epsilon \approx 0.83281573$.

7. $7! = 4^2 3^2 5 \times 7 = 1^2 12^2(5^2 + 10)$, $a = 12$, $a_1 = 1$, $b = 35$, $c = 5$, $d = 10$, $\epsilon_3 \approx 0.916079783$, $k = 70$, $\epsilon \approx 0.992957397$.

8. $8! = 8^2 3^2 2 \times 5 \times 7 = 3^2 8^2(8^2 + 6)$, $a = 24$, $a_1 = 3$, $b = 70$, $c = 8$, $d = 6$, $\epsilon_3 \approx 0.366600265$, $k = 200$, $\epsilon \approx 0.798406368$.

9. $9! = 8^2 9^2 2 \times 5 \times 7 = 8^2 9^2(8^2 + 6)$, $a = 72$, $a_1 = 8$, $b = 70$, $c = 8$, $d = 6$, $\epsilon_3 \approx 0.366600265$, $k = 602$, $\epsilon \approx 0.39521908$.

10. $10! = 16^2 9^2 5^2 7 = 72^2 10^2(2^2 + 3)$, $a = 720$, $a_1 = 72$, $b = 7$, $c = 2$, $d = 3$, $\epsilon_3 \approx 0.645751311$, $k = 1904$, $\epsilon \approx 0.940943967$.

11. $11! = 16^2 9^2 5^2 7 \times 11 = 60^2 12^2(8^2 + 13)$, $a = 720$, $a_1 = 60$, $b = 77$, $c = 8$, $d = 13$, $\epsilon_3 \approx 0.774964387$, $k = 6317$, $\epsilon \approx 0.974358922$.

12. $12! = 32^2 9^2 5^2 3 \times 7 \times 11 = 120^2 12^2(8^2 + 13)$, $a = 1440$, $a_1 = 120$, $b = 231$, $c = 15$, $d = 6$, $\epsilon_3 \approx 0.198684154$, $k = 21886$, $\epsilon \approx 0.105181142$.

13. $13! = 32^2 9^2 5^2 3 \times 7 \times 11 \times 13 = 96^2 15^2(8^2 + 13),\ a = 1440,\ a_1 = 96,\ b = 3003,\ c = 54,\ d = 87,\ \epsilon_3 \approx 0.799635035,\ k = 83517,\ \epsilon \approx 0.4744504.$

14. $14! = 32^2 9^2 5^2 7^2 2 \times 3 \times 11 \times 13 = 720^2 14^2(29^2 + 17),\ a = 10080,\ a_1 = 720,\ b = 858,\ c = 29,\ d = 17,\ \epsilon_3 \approx 0.291637032,\ k = 295259,\ \epsilon \approx 0.701280076.$

15. $15! = 32^2 27^2 5^2 7^2 2 \times 5 \times 11 \times 13 = 2016^2 15^2(37^2 + 61),\ a = 30240,\ a_1 = 2016,\ b = 1430,\ c = 37,\ d = 61,\ \epsilon_3 \approx 0.815340802,\ k = 1143535,\ \epsilon \approx 0.905863913.$

16. $16! = 128^2 27^2 5^2 7^2 2 \times 5 \times 11 \times 13 = 7560^2 16^2(37^2 + 61),\ a = 120960,\ a_1 = 7560,\ b = 1430,\ c = 37,\ d = 61,\ \epsilon_3 \approx 0.815340802,\ k = 4574143,\ \epsilon \approx 0.623455652.$

17. $17! = 128^2 27^2 5^2 7^2 2 \times 5 \times 11 \times 13 \times 17 = 6720^2 18^2(155^2 + 285),\ a = 120960,\ a_1 = 6720,\ b = 24310,\ c = 155,\ d = 285,\ \epsilon_3 \approx 0.815340802,\ k = 18859677,\ \epsilon \approx 0.306253148.$

18. $18! = 256^2 81^2 5^2 7^2 5 \times 11 \times 13 \times 17 = 40320^2 18^2(110^2 + 55),\ a = 725760,\ a_1 = 40320,\ b = 12155,\ c = 110,\ d = 55,\ \epsilon_3 \approx 0.249716553,\ k = 80014834,\ \epsilon \approx 0.28550528.$

19. $19! = 256^2 81^2 5^2 7^2 5 \times 11 \times 13 \times 17 \times 19 = 36288^2 20^2(480^2 + 545),\ a = 725760,\ a_1 = 36288,\ b = 230945,\ c = 480,\ d = 545,\ \epsilon_3 \approx 0.567373008,\ k = 348776576,\ \epsilon \approx 0.63428608.$

20. $20! = 512^2 81^2 25^2 7^2 11 \times 13 \times 17 \times 19 = 362880^2 20^2(214^2 + 393),\ a = 7257600,\ a_1 = 362880,\ b = 46189,\ c = 214,\ d = 393,\ \epsilon_3 \approx 0.916262763,\ k = 1559776268,\ \epsilon \approx 0.6287488.$

21. $21! = 512^2 81^2 25^2 7^2 3 \times 7 \times 11 \times 13 \times 17 \times 19 = 345600^2 21^2(984^2 + 1713),\ a = 7257600,\ a_1 = 345600,\ b = 969969,\ c = 984,\ d = 1713,\ \epsilon_3 \approx 0.870042188,\ k = 7147792818,\ \epsilon \approx 0.1836288.$

22. $22! = 512^2 81^2 25^2 7^2 11^2 2 \times 3 \times 7 \times 13 \times 17 \times 19 = 3628800^2 22^2(419^2 + 797),\ a = 79833600,\ a_1 = 3628800,\ b = 176358,\ c = 419,\ d = 797,\ \epsilon_3 \approx 0.949997023,\ k = 33526120082,\ \epsilon \approx 0.3353728.$

23. $23! = 512^2 81^2 25^2 7^2 11^2 2 \times 3 \times 7 \times 13 \times 17 \times 19 \times 23 = 3326400^2 24^2(2014^2 + 38),\ a = 79833600,\ a_1 = 3326400,\ b = 4056234,\ c = 2014,\ d = 38,\ \epsilon_3 \approx 0.00943394,\ k = 160785623545,\ \epsilon \approx 0.392384.$

24. $24! = 2048^2 243^2 25^2 7^2 11^2 7 \times 13 \times 17 \times 19 \times 23 = 79833600^2 24^2(822^2 + 355),\ a = 1916006400,\ a_1 = 79833600,\ b = 676039,\ c = 822,\ d = 355,\ \epsilon_3 \approx 0.2159083844,\ k =,\ \epsilon \approx 0.5576576.$

2.12.2 Lemma

For a given natural number $n > 1$, we know $n! = a^2 b = a^2(c^2 + d),\ d \leq 2c,\ c < an$ and $n + 1 = c_1^2 + d_1,\ d_1 \leq 2c_1$. Then $(n + 1)! < a^2(3c^2 c_1^2 + 2cc_1),\ c_1 < \sqrt{n}$.

Proof: Given that $n! = a^2b = a^2(c^2+d),\ d \le 2c,\ c < an$ and $n+1 = c_1^2 + d_1,\ d_1 \le 2c_1$. Then for $d = 2c,\ d_1 = 2c_1;\ (n+1)! = a^2(c^2+2c)(c_1^2+2c_1) = a^2(c^2c_1^2 + 2cc_1^2 + 2c_1c^2 + 4cc_1) = a^2(c^2c_1^2 + 2cc_1(c+c_1+2)) < a^2(3c^2c_1^2 + 2cc_1)$ as $1+c+c_1 < c+c_1$ when $cc_1 - c - c_1 + 1 > 2$ i.e. $(c-1)(c_1-1) > 2$. Again $n+1 = c^2 + 2c_1$ gives $c_1 < \sqrt{n}$ and $\sqrt{3}\ cc_1 < \sqrt{3}\sqrt{n}\ an < n^2a$.

2.13 Theorem

If $n! = a^2b = a^2(c^2+d),\ d \le 2c,\ n \in \mathbb{N}$ and $\sqrt{n!} = k+\epsilon$, then $2k\epsilon+\epsilon^2 = a^2d - [a\epsilon_3](2k - [a\epsilon_3]) = a^2d - [a\epsilon_3](2ac + [a\epsilon_3]) = n! - k^2$, where $\sqrt{c^2+d} = c+\epsilon_3$. For the solution of Brocard-Ramanujan problem (c.f. (1)), $2k\epsilon + \epsilon^2 = a^2d - [a\epsilon_3](2k - [a\epsilon_3]) = a^2d - [a\epsilon_3](2ac + [a\epsilon_3]) = 2k$ and $k = \frac{a^2d + [a\epsilon_3]^2}{2(1+[a\epsilon_3])}$.

Proof: For natural number $n(> 1)$, we know that

$$n! = a^2(c^2+d),\ d \le 2c. \tag{24}$$

$$\text{Then } \sqrt{n!} = a(c+\epsilon_3) \text{ (where } 2c\epsilon_3 + \epsilon_3^2 = d) \tag{25}$$

$$= (ac + [a\epsilon_3]) + (a\epsilon_3 - [a\epsilon_3]) = k + \epsilon,\ k = ac + [a\epsilon_3],\ \epsilon = a\epsilon_3 - [a\epsilon_3]. \tag{26}$$

Now, $2k\epsilon + \epsilon^2 = 2(ac + [a\epsilon_3])(a\epsilon_3 - [a\epsilon_3]) + (a\epsilon_3 - [a\epsilon_3])^2 = a\epsilon_3(2ac + a\epsilon_3) - [a\epsilon_3](2ac + [a\epsilon_3])$

$$= a^2(2c\epsilon_3 + \epsilon_3^2) - [a\epsilon_3](2ac + [a\epsilon_3])$$

$$= a^2d - [a\epsilon_3](2ac + [a\epsilon_3]) = a^2d - [a\epsilon_3](2k - [a\epsilon_3]) \tag{27}$$

$$= a^2d - (k - ac)(k + ac) = a^2d - k^2 + a^2c^2 = a^2(c^2+d) - k^2 = n! - k^2. \tag{28}$$

Moreover, we know that $n! = k^2 + 2k\epsilon + \epsilon^2$ i.e. $2k\epsilon + \epsilon^2 = n! - k^2$. Thus for the solution of Brocard-Ramanujan problem, we have from (27)

$$2k\epsilon + \epsilon^2 = a^2d - [a\epsilon_3](2ac + [a\epsilon_3]) = a^2d - [a\epsilon_3](2k - [a\epsilon_3]) = 2k. \tag{29}$$

$$\text{Then } k = ac + [a\epsilon_3] = \frac{a^2d + [a\epsilon_3]^2}{2(1 + [a\epsilon_3])} \tag{30}$$

For the solution of Brocard-Ramanujan problem, we have the following examples:

1. $4! = 2^26 = 2^2(2^2+2),\ a = 2,\ c = 2,\ d = 2,\ \ \epsilon_3 \approx 0.449489743,\ k = 4,\ \epsilon \approx 0.898979486.$

2. $5! = 2^2 \times 2 \times 3 \times 5 = 2^2(5^2+5),\ a = 2,\ b = 30,\ c = 5,\ d = 5,\ \epsilon_3 \approx 0.477225575,\ k = 10,\ \epsilon \approx 0.95445115.$

3. $7! = 4^23^25 \times 7 = 1^212^2(5^2+10),\ a = 12,\ b = 35,\ c = 5,\ d = 10,\ \epsilon_3 \approx 0.916079783,\ k = 70,\ \epsilon \approx 0.992957397.$

2.13.1 Remark

We know that k is strictly increasing (if $n > 2$) and a is increasing with the increase value of n. Thus from (30), k will be increasing if $a,\ d$ (and hence may be $[a\epsilon_3]$) are increasing. We can observe that $d = c$ for first and second solution of Brocard-Ramanujan problem. For the third solution, it is $d = 2c$. Also, there is no other solution for $n \leq 10^9$. Moreover, $n! = k^2 + 2k$ for the solution and $n! < k^2 + 2k$ for non-solution of (1). Thus $a,\ d,\ \epsilon_3$ are increasing for maximum increase of k to the solution, however maximum value of d is $2c$. Hence, we may claim here also that there will be no other solution of (1).

2.13.2 Corollary

$n! = k^2 + a^2d - [a\epsilon_3](2ac + [a\epsilon_3]) = k^2 - [a\epsilon_3](2k - [a\epsilon_3]) = k^2 + (n! - k^2)$.

2.14 Theorem

If $n! = a^2b = a^2(c^2 + d),\ d \leq 2c,\ n \in \mathbb{N}$, then $n! + 1 = (ac + e)^2$ if $a^2d + 1 = e(2ac + e)$ for the solution of Brocard-Ramanujan problem and $(ac + e)^2 - 1 < n! + 1 < (ac + e)^2$ for the non-solution of (1). By division algorithm $a^2d + 1 = q \times 2ac + r,\ r < 2ac$. Then (1) has solution if and only if $a^2c^2 + 2acq + r =$ square. Moreover, the value of e is $e = q - [x]$ where $x^2 - 2(ac + q)x + (q^2 - r) = 0$.

Proof: We have

$$a^2d + 1 = q \times 2ac + r,\ (r < 2ac). \tag{31}$$

$$= (q - x)\{2ac + (q - x)\} + r + 2acx - (q - x)^2. \tag{32}$$

$$\text{Then } n! + 1 = (ac + q - x)^2 \text{ if } 2acx - (q - x)^2 + r = 0. \tag{33}$$

$$\text{Or, } x^2 - 2(ac + q)x + (q^2 - r) = 0. \tag{34}$$

$$\text{Then, } x = \frac{2(ac + q) \pm \sqrt{4(ac + q)^2 - 4(q^2 - r)}}{2} = (ac + q) \pm \sqrt{a^2c^2 + 2acq + r}. \tag{35}$$

For the solution of (1), we need $x \in \mathbb{N}$.

$$\text{Then, } a^2c^2 + 2acq + r = \text{square} = f^2,\ f \in \mathbb{N}; \tag{36}$$

$$n! + 1 = (ac + e)^2 = f^2 = m^2,\ e = q - x,\ f = ac + (q - x). \tag{37}$$

For the non-solution of (1), we have $x \notin \mathbb{N}$. Then $x = (ac + q) - \sqrt{a^2c^2 + 2acq + r} = x_1 + x_2,\ 0 < x_2 < 1$; $(ac+e)^2 - 1 < n! + 1 < (ac+e)^2$, $e = q - x_1 = q - [x]$. Moreover, $a^2d + 1 =$

$(q-x)\{2ac+(q-x)\}$ as $r+2acx-(q-x)^2=0$. So $a^2d+1=(q-x_1-x_2)\{2ac+(q-x_1-x_2)\}=(q-x_1)\{2ac+(q-x_1)\}-2acx_2-2x_2(q-x_1)+x_2^2$. Thus $acx_2+2x_2(q-x_1)-x_2^2<2(ac+e)-1$. Some examples of this theorem have been presented below.

1. $2!=1^2(1^1+1),\ a=1,\ c=1,\ d=1,\ a^2d+1=2=1\times2+0,\ x=2-\sqrt{3},\ e=q-[x]=1,\ ac+e=2,\ 1^2<2!+1<2^2$.

2. $3!=1^2(2^2+2),\ a=1,\ c=2,\ d=2,\ a^2d+1=3=0\times4+3,\ x=2-\sqrt{7},\ e=q-[x]=1,\ ac+e=3,\ 2^2<3!+1<3^2$.

3. $4!=2^2(2^2+2),\ a=2,\ c=2,\ d=2,\ a^2d+1=9=1\times8+1,\ x=5-\sqrt{5^2},\ e=q-x=1,\ ac+e=5,\ 4!+1=5^2$.

4. $5!=2^2(5^2+5),\ a=2,\ c=5,\ d=5,\ a^2d+1=21=1\times20+1,\ x=11-\sqrt{11^2},\ e=q-x=1,\ ac+e=11,\ 5!+1=11^2$.

5. $6!=12^2(2^2+1),\ a=12,\ c=2,\ d=1,\ a^2d+1=145=3\times48+1,\ x=27-\sqrt{27^2-8},\ e=q-[x]=3,\ ac+e=27,\ 26^2<6!+1<27^2$.

6. $7!=12^2(5^2+10),\ a=12,\ c=5,\ d=10,\ a^2d+1=1441=12\times120+1,\ x=72-\sqrt{71^2},\ e=q-x=11,\ ac+e=71,\ 7!+1=71^2$.

7. $8!=24^2(8^2+6),\ a=24,\ c=8,\ d=6,\ a^2d+1=3457=9\times384+1,\ x=201-\sqrt{200^2+321},\ e=q-[x]=9,\ ac+e=201,\ 200^2<8!+1<201^2$.

8. $17!=120960^2(155^2+285),\ a=120960,\ c=155,\ d=285,\ a^2d+1=3=111205\times37497600+6048001,\ x=18860005-\sqrt{120960^2155^2+2\times120960\times155\times6048001}\approx327.693746824,\ e=q-[x]=110878,\ ac+e=18859678,\ 18859677^2<17!+1<18859678^2$.

2.14.1 Corollary

If $d=2c$, then $a^2c^2+2a^2c+1=f^2,\ f^2-1=a^2(f_1^2-1);\ f,\ f_1\in\mathbb{N}$ for the solution of (1).

Proof:

$$\text{For } d=2c,\ a^2d+1=2a^2c+1=a\times2ac+1,\ q=a,\ r=1. \tag{38}$$

$$x=(ac+a)-\sqrt{a^2c^2+2a^2c+1}. \tag{39}$$

For the solution of (1), we need $x\in\mathbb{N}$.

$$\text{Then } a^2c^2+2a^2c+1=f^2,\ f\in\mathbb{N}. \tag{40}$$

$$\text{Or, } c=\frac{-2a^2+\sqrt{4a^4+4a^2(f^2-1)}}{2a^2}=-1+\frac{\sqrt{a^2+f^2-1}}{a}. \tag{41}$$

Thus for the solution of (1), a^2 divides f^2-1,

$$f^2-1=a^2(f_1^2-1),\ f_1\in\mathbb{N} \tag{42}$$

$$\text{and } x=(ac+a)-f,\ c=-1+f_1. \tag{43}$$

When $n=7,\ a=12,\ 7!=12^2 35=12^2(5^2+2\times 5)=12^2(6^2-1)=71^2-1,\ f=71,\ f_1=6.$ Moreover, $5040=71^2-1=70\times 72=12^2(6^2-1)=60\times 84$. Thus $7!+1=71^2$.

We also can find that $f=97,\ f^2-1=96\times 98=14^2(7^2-1)=14^2(6^2+2\times 6)$. But there does not exist $n\in\mathbb{N}$ such that $n!=14^2(6^2+12)$. We can generate infinite number of such solutions of $f^2-1=a^2(f_1^2-1)$ which follow unique pattern as given below.

$4^2(1^2+2\times 1)=4^2(2^2-1)=7^2-1,$

$6^2(2^2+2\times 2)=6^2(3^2-1)=17^2-1,$

$8^2(3^2+2\times 3)=8^2(4^2-1)=31^2-1,$

$10^2(4^2+2\times 4)=10^2(5^2-1)=49^2-1$

$16^2(7^2+2\times 7)=16^2(8^2-1)=127^2-1,$

$18^2(8^2+2\times 8)=18^2(9^2-1)=161^2-1,$

$20^2(9^2+2\times 9)=20^2(10^2-1)=199^2-1.$

Thus we have the unique pattern as

$$(2c+2)^2(c^2+2c)=(2c+2)^2\{(c+1)^2-1\}=\{2c(c+2)+1\}^2-1,\ c\in\mathbb{N}. \tag{44}$$

Also from (39), (40) and (44) using $a=2c+2$, we can obtain

$$x=(c+1)(2c+2)-\sqrt{2^2c^2(c+2)^2+4c(c+2)+1}=2(c+1)^2-2c(c+2)=1. \tag{45}$$

Analyzing the section 2.12.1 Remark, it can be discovered that a is even for $n>3$, $2c+2<a$ for $n>7$ and $nc<<a$ for large n ($n\geq 25$). Hence, there will be no other solution of (1) following pattern of equation (44) except the only solution $7!+1=71^2$ when $d=2c$. However, $a\leq c$ when $n\leq 5$. Then for the solution of (1), we need $d<2c$. We have already observed that there are two solutions $4!+1=5^2$ and $5!+1=11^2$ of (1) when $d=c$.

2.14.2 Corollary

If $d=c$, then $a^2c^2+a^2c+1=f^2,\ 4(f^2-1)=a^2(f_1^2-1);\ f,\ f_1\in\mathbb{N}$ for the solution of (1).

Proof:

$$\text{For } d=c,\ a^2d+1=a^2c+1=\frac{a}{2}\times 2ac+1,\ q=\frac{a}{2},\ r=1. \tag{46}$$

$$x=(ac+a)-\sqrt{a^2c^2+2a^2c+1}. \tag{47}$$

For the solution of (1), we need $x \in \mathbb{N}$.

$$\text{Then } a^2c^2 + a^2c + 1 = f^2,\ f \in \mathbb{N}. \tag{48}$$

$$\text{Or, } c = \frac{-a^2 + \sqrt{a^4 + 4a^2(f^2-1)}}{2a^2} = -\frac{1}{2} + \frac{\sqrt{a^2 + 4(f^2-1)}}{2a}. \tag{49}$$

Thus for the solution of (1), a^2 divides $4(f^2-1)$,

$$4(f^2-1) = a^2(f_1^2-1),\ f_1 \in \mathbb{N} \tag{50}$$

$$\text{and } x = \left(ac + \frac{a}{2}\right) - f,\ c = -\frac{1}{2} + \frac{f_1}{2},\ f_1 \text{ is odd.} \tag{51}$$

When $n = 4,\ a = 2,\ 4! = 2^2 6 = 2^2(2^2+2) = 5^2 - 1,\ 2^2(f^2-1) = a^2(f_1^2-1),\ f = f_1 = 5$; and for $n = 5,\ a = 2,\ 5! = 2^2 30 = 2^2(5^2+5) = 11^2 - 1,\ 2^2(f^2-1) = a^2(f_1^2-1),\ f = f_1 = 11$.

We also can find that $2^2(6^2+6) = 13^2 - 1,\ a = 2,\ f = f_1 = 13$. But there does not exist $n \in \mathbb{N}$ such that $n! = 2^2(6^2+6)$. We can generate infinite number of such solutions of $2^2(f^2-1) = a^2(f_1^2-1)$ which follow unique pattern as given below.

$2^2(1^2+1) = 3^2 - 1,\ a = 2,\ f = f_1 = 3,$

$2^2(3^2+3) = 7^2 - 1,\ a = 2,\ f = f_1 = 7,$

$2^2(4^2+4) = 7^2 - 1,\ a = 2,\ f = f_1 = 9,$

$2^2(7^2+7) = 15^2 - 1,\ a = 2,\ f = f_1 = 15,$

$2^2(8^2+8) = 17^2 - 1,\ a = 2,\ f = f_1 = 17,$

$2^2(9^2+9) = 19^2 - 1,\ a = 2,\ f = f_1 = 19.$

Thus we have the unique pattern as

$$2^2(c^2+c) = (4c^2+4c) = (2c+1)^2 - 1, a = 2,\ f = f_1 = 2c+1,\ c \in \mathbb{N}. \tag{52}$$

Analyzing the section 2.12.1 Remark, it can be discovered that a is even for $n > 3$, $a > 2c+2$ for $n > 7$ and $a >> nc$ for large n ($n > 50$). Hence, there will be no other solution of (1) following pattern of equation (52) except the two solutions $4! + 1 = 5^2$ and $5! + 1 = 11^2$ when $d = c$.

2.14.3 Corollary

By further thoroughly investigation of the pattern of (44), we can generate infinite number of similar patterns.

Proof: Following the further investigation of the pattern of (44), we can get

$$\{2(2c^2+4c)+2\}^2\{(2c^2+4c)^2 + 2(2c^2+4c)\}$$

$$= \{8(c^2+2c)^2 + 8(c^2+4c) + 2\}\{8(c^2+2c)^2 + 8(c^2+4c)\}$$

$$\text{Or, } 4^2\{2c(c+2)+1\}^2(c+1)^2(c^2+2c)\} = \{8c(c+1)^2(c+2)+1\}^2 - 1. \tag{53}$$

Here $a = 4(c+1)\{2c(c+2)+1\},\ d = 2c,\ q = a,\ r = 1$ and then

$$x = (ac+q) - \sqrt{a^2c^2 + 2acq + r}$$
$$= 4(c+1)^2\{2c(c+2)+1\} - \sqrt{64c^8 + 512c^7 + 1664c^6 + 2816c^5 + 2640c^4 + 1344c^3 + 336c^2 + 32c + 1}$$
$$= 4(c-1)(c+3) + 15. \tag{54}$$

Following (53), we again obtain

$$\{16c(c+1)^2(c+2)+2\}^2\{8c(c+1)^2(c+2)\}\{8c(c+1)^2(c+2)+2\}$$
$$= \{128c^2(c+1)^4(c+2)^2 + 32c(c+1)^2(c+2)+2\}\{128c^2(c+1)^4(c+2)^2 + 32c(c+1)^2(c+2)\} \tag{55}$$

$$\text{Or, } 8^2\{8c(c+1)^2(c+2)+1\}^2\{2c(c+2)+1\}^2(c+1)^2(c^2+2c)$$
$$= \{32c(c+1)^2(c+2)\{2c(c+2)+1\}^2+1\}^2 - 1. \tag{56}$$

We can repeat this process.

2.14.4 Corollary

Investigating the pattern of (44), we can formulate infinite number of similar patterns for $a^2(c^2+d)$.

Proof: Following the pattern of (44), we can evaluate

$$\{2(2c+1)\}^2(4c^2+4c) = (8c^2+8c+1)(8c^2+8c). \tag{57}$$
$$\text{Or, } 4^2(2c+1)^2(c^2+c) = \{8c(c+1)+1\}^2 - 1. \tag{58}$$

Here $a = 4(2c+1),\ d = c,\ q = \frac{a}{2},\ r = 1$ and then

$$x = (ac+q) - \sqrt{a^2c^2 + 2acq + r} = 2(2c+1)^2 - \sqrt{64c^4 + 128c^3 + 80c^2 + 16c + 1} = 1. \tag{59}$$

Following (58), we again obtain

$$\{16c(c+1)^2(c+2)+2\}^2\{8c(c+1)^2(c+2)\}\{8c(c+1)^2(c+2)+2\}$$
$$= \{128c^2(c+1)^4(c+2)^2 + 32c(c+1)^2(c+2)+2\}\{128c^2(c+1)^4(c+2)^2 + 32c(c+1)^2(c+2)\} \tag{60}$$

$$\text{Or, } 8^2\{8c(c+1)^2(c+2)+1\}^2\{2c(c+2)+1\}^2(c+1)^2(c^2+2c)$$
$$= \{32c(c+1)^2(c+2)\{2c(c+2)+1\}^2+1\}^2 - 1. \tag{61}$$

We can also repeat this process.

2.14.5 Corollary

Investigating the pattern of (44), we can formulate infinite number of similar patterns for $a^2(c^2+1)$.

Proof: Following the pattern of (44), we have

$$4c^2(c^2+1) = (2c^2+1)^2 - 1. \tag{62}$$

Again repeating this process, we can also obtain

$$16c^2(2c^2+1)^2(c^2+1) = \{8c^2(c^2+1)\}^2 - 1. \tag{63}$$

2.14.6 Corollary

Investigating the pattern of (44), we can obtain infinite number of similar patterns for $a^2(c^2+2)$.

Proof: The pattern of (44) helps us to formulate

$$c^2(c^2+2) = (c^2+1)^2 - 1. \tag{64}$$

Repeating this process, we can also obtain

$$4c^2(c^2+1)^2(c^2+2) = \{2c^2(c^2+2)+1\}^2 - 1. \tag{65}$$

2.14.7 Note

The patterns in sections 2.14.3-2.14.6 do not provide any new solution of Brocard-Ramanujan problem.

2.15 Theorem

We can claim that Brocard-Ramanujan problem has no further solution using regular growth of k and uncertain growth of ϵ.

Proof: By theorem (2.3) and corollary (2.3.2), we know that for the solution of Brocard-Ramanujan problem (1); $m-1 = k = [\sqrt{n!}]$, $\sqrt{n!} = k+\epsilon$, where $\epsilon = \sqrt{n!} - [\sqrt{n!}]$ i.e. $0 < \epsilon < 1$. Then for the solution of (1), the natural number $\epsilon(2k+\epsilon)$ must satisfy $\epsilon(2k+\epsilon) = 2k$; $k = \frac{\epsilon^2}{2(1-\epsilon)}$ and $\epsilon = \sqrt{k^2+2k} - k$ should be distinct and monotonic increasing for all the solutions of (1).

Moreover, for any natural number n, $[\sqrt{n!}] = k$. Then for the natural number $n+1$, we have $[\sqrt{(n+1)!}] = [k\sqrt{n+1}] \geq k[\sqrt{n+1}]$. Thus for every growth of n, k is growing with

multiplicity $\sqrt{n+1}$. However, ϵ may increase or decrease as shown with examples in remark 2.12.1. Also, both the growths should satisfy $k = \frac{\epsilon^2}{2(1-\epsilon)}$ and $\epsilon = \sqrt{k^2+2k} - k$.

According to Bruce Berndt and William Galway [10], there is no new solution of the problem based on computations up to $n = 10^9$. Thus for $n \geq 10^9$, we have huge growth of k by regular growth with multiplicity $\sqrt{n+1}$, whereas growth of ϵ is uncertain such that its growth can not match the growth of k to satisfy Brocard-Ramanujan problem (1). Hence, we can claim that Brocard-Ramanujan problem has no further solution.

2.16 Conclusion

The study explores multiple results on Brocard-Ramanujan problem (1). For natural number n, if $\sqrt{n!} = k + \epsilon,\ n > 1,\ 0 < \epsilon < 1$; then it has solution if and only if $n! = k(k+2)$ and $\epsilon,\ k$ are strictly monotonic increasing. The problem has only finitely many solutions which is not based on any conjecture or previous study on the Brocard-Ramanujan problem. For the new solution when $n \geq 10^5$, the value of ϵ should be more than $0.999\cdots905915$ (digit 0 is coming after 228287 numbers of 9 digit taking more than 66 pages in LibreOffice Writer) indicating almost impossibility of new solution. For $n \geq 10^9$, I am unable to calculate the said numbers of 9 digit in the value of ϵ in my personal laptop (with 8GB Ram) using MATHEMATICA 8.

For the solution of (1), the natural number $\epsilon(2k+\epsilon)$ must satisfy $\epsilon(2k+\epsilon) = 2k$ and conversely. Moreover, $k = \frac{\epsilon^2}{2(1-\epsilon)}$ and $\epsilon = \sqrt{k^2+2k} - k$ should be distinct and monotonic increasing for all the solutions of (1) and $n! < k(k+2)$ if n is not a solution of Brocard-Ramanujan problem (c.f. equation (1)).

If $\sqrt{n!} = k + \epsilon$ for solution of Brocard-Ramanujan problem (c.f. equation (1)), then (ϵ, k) lies on hyperbola.

If $n! = a^2b = a^2(c^2+d),\ d \leq 2c,\ n \in \mathbb{N},\ \sqrt{n!} = k+\epsilon,\ \sqrt{c^2+d} = c + \epsilon_3$; then for the solution of Brocard-Ramanujan problem, $k = \frac{a^2d+[a\epsilon_3]^2}{2(1+[a\epsilon_3])}$. Moreover, if $n!+1 = (ac+e)^2$, then $a^2d+1 = e(2ac+e)$ for the solution of Brocard-Ramanujan problem and $(ac+e)^2 - 1 < n! + 1 < (ac+e)^2$ for the non-solution of (1). By division algorithm $a^2d+1 = q \times 2ac + r,\ r < 2ac$ and (1) has solution if and only if $a^2c^2 + 2acq + r =$ square. Also, the value of e is $e = q - [x]$ where $x^2 - 2(ac+q)x + (q^2 - r) = 0$.

Finally, using the results of sections 2.3, 2.3.1, 2.10, 2.13.1, 2.13, 2.14,2.14.1,2.14.2 2.15; it has been claimed to discover that the problem has no further solution.

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